

LUYAO NIU (SHE/HER)

PhD student, University of Southern California
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EDUCATION

University of Southern California

PhD · Transportation Systems Engineering | 2026.08–present

Peking University

M.Sc. · Urban Planning / Smart City & Big Data | 2023–2026

Beijing Jiaotong University

B.Sc. · Traffic Engineering; minor in Finance | 2019–2023

RESEARCH INTERESTS

Learning, decision-making, and coordination in intelligent systems, with a growing focus on multi-agent and autonomous systems.

Learning & Reasoning | Multi-Agent Decision-Making | Coordination & Collective Intelligence

Current topic at USC: Coalition coordination in mixed-autonomy traffic

When should autonomous-vehicle firms route independently, coordinate in partial coalitions, or share a common dispatcher? I study how these organizational choices affect traffic equilibrium and network performance when human drivers also adapt their routes.

SELECTED RESEARCH EXPERIENCE

Coalition Coordination in Mixed-Autonomy Traffic |

- Question: Studies when autonomous-vehicle firms should route independently, coordinate in partial coalitions, or share a common dispatcher when human drivers also adapt their routes.
- Approach: Models heterogeneous routing organizations through mixed Nash-Wardrop equilibrium and compares their operational implications under fixed network and demand conditions.

Event-CausNet: Reliable Spatio-Temporal Forecasting |

- Question: Examines how unstructured event reports can improve traffic forecasts during non-recurring events.
- Approach: Uses language models to extract event factors and combines causal knowledge with spatio-temporal forecasting. Submitted to KDD 2027.

AskNearby: Community Information Retrieval and Recommendation |

- Question: Addresses local information accessibility by combining neighborhood knowledge with individual context and preferences.
- Approach: Built a multi-layer retrieval pipeline spanning knowledge-graph retrieval, semantic-vector recall, and geographic filtering. ACM SIGSPATIAL GeoAI 2025 oral presentation.

PROFESSIONAL EXPERIENCE

AI Agent Engineering Intern | May - Aug. 2026

Xiaohongshu · Data Analytics

- Domain knowledge infrastructure: Built a domain-specific data analytics knowledge base to support knowledge-grounded agent workflows.
- Agent architecture and development: Developed agent frameworks and data-analysis workflows that connect knowledge resources with domain capabilities.
- Validation and Skill operations: Validated end-to-end agent behavior and analytical outputs; designed, evaluated, monitored, and operated reusable domain Skills.

Co-founder | Dec. 2024 - Present

Qianmo Zhilian (Shenzhen) Technology Co., Ltd.

- AskNearby: Co-founded and led development of an LLM-powered community information platform combining RAG and spatio-temporal data.
- Recognition: The project received a Gold Award at the Nanshan District Innovation & Entrepreneurship Competition and Third Prize at the Baidu ERNIE Cup.

Research Assistant | Jan. 2025 - Present

Data Science Department, Duke Kunshan University · Suzhou, China

- Research leadership: Led independent research on spatio-temporal data mining, traffic forecasting, and behavioral modeling.
- Research output: First-authored work published at IEEE GLOBECOM; Event-CausNet submitted to KDD 2027.

LLM Algorithm R&D | Apr. 2025 - Jan. 2026

China Academy of Land Surveying and Planning · Collaborative project

- Knowledge and reporting agents: Led the RAG retrieval module for a policy knowledge base and developed an LLM-agent reporting workflow.
- Workflow impact: The reporting workflow reduced initial-draft preparation time by more than 90%.

AWARDS & ACADEMIC SERVICE

Scholarships & honors

- Baosteel Outstanding Student Award (500 recipients nationwide / year), 2022.12
- National Scholarship (highest undergraduate honor in China, top 1%), 2020.12
- First-class Academic Scholarship (top 3%), 2020.12
- Beijing Outstanding Graduate

Competition awards

- First Prize, National University Traffic Science & Technology Competition, 2022.07
- Second Prize, China MathorCup Mathematical Modeling Competition, 2022.04
- Third Prize, China College Mathematics Competition, 2020.04
- Third Prize, Baidu ERNIE Cup AI Entrepreneurship Competition, 2024.09
- Gold Award, Nanshan District Innovation & Entrepreneurship Competition

Academic Service

Peer Reviewer: WSDM · IEEE GLOBECOM · Transportation Research Board (TRB)

LEADERSHIP & SERVICE

President, Student Union | Beijing Jiaotong University | Jun. 2021 - Jun. 2023

Led university-wide student affairs and brand events reaching 80,000+ participants.

Volunteer, Beijing 2022 Winter Olympics | National Indoor Stadium | Jan. 2022 - Apr. 2022

Personnel management and venue operations; named Outstanding Volunteer.

PUBLICATIONS

Journal

Niu, C.; He, S.; Fang, C.; **Niu, L.**; Xiao, Y.; Zhang, W. (2026). Gender and Age Disparities in Public Health Crisis Disruptions to Cross-Border Mobility: A Causal Analysis of Daily Flows Between Shenzhen and Hong Kong. *Systems*, 14(7), 772. <https://doi.org/10.3390/systems14070772>

Shi, Z.; Luo, S.; **Niu, L.**; Shao, C. (2025). Joint Design of Public Transit and Autonomous Vehicle Mobility Systems for Tourist Cities Considering Abnormal Trips. *Transportation Research Part C*. <https://doi.org/10.1016/j.trc.2025.105179>

Liu, C.; **Niu, L.**; Zhang, W. (2025). Rebound or Substitution? The Causal Effect of Electric Vehicle Adoption on Individual Travel Behaviour: A Quasi-Experiment. *Applied Geography* (Under review).

Conference

Niu, L.; Deng, Z.; Li, B.; Huang, N.; Liu, R.; Zhang, W. (2025). AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations. *ACM SIGSPATIAL GeoAI '25*. <https://doi.org/10.48550/arXiv.2512.02502>

Niu, L.; Wang, Z.; Liu, J.; Boukerche, A.; Sun, P. (2025). MF-AttnBiLSTM: Traffic Flow Prediction via Hybrid Signal Decomposition and Dual-Stream Temporal Attention Learning. *IEEE GLOBECOM 2025*.

Niu, L.; Wang, Y.; He, S.; Zhang, W. (2025). Evaluating the Causal Impacts on Spatial Satisfaction Sentiment of Shenzhen–Hong Kong Regional Integration Policies: An Interrupted Time Series Study Leveraging LLMs. *EUGEO 2025*.

Niu, L.; Gu, J.; Liu, C.; Zhang, W. (2025). Why Public Transport Policies Work or Not? A Meta-analysis Based on the Behaviour Change Wheel. *AESOP 2025*.

Wang, Z.*; **Niu, L.***; Ban, J. (2027). GeoSplit: Selective Map Acquisition for Unseen-User Travel-Mode Recognition. *ICASSP 2027* (Submitted).

* *Equal contribution.*

Niu, L.; Wang, Z. (2027). Whose Normal? Reference and Perturbation Controls for Personal Mobility Anomaly Detection. *ICASSP 2027* (Submitted).

Manuscripts & Preprints

Niu, L.; Wang, Z.; Guan, S.; Sun, P. (2025). Event-CausNet: Unlocking Causal Knowledge from Text with LLMs for Reliable Spatio-Temporal Forecasting. *KDD 2027* (Submitted). <https://doi.org/10.48550/arXiv.2511.12769>

Niu, L.; Huang, N. (2025). ST-ProC: A Graph-Prototypical Framework for Robust Semi-Supervised Travel Mode Identification. *arXiv* (Preprint). <https://doi.org/10.48550/arXiv.2511.13702>

Niu, L.; Wang, Y.; He, S.; Zhang, W. (2025). Leveraging Large Language Models to Measure Cross-Border Spatial Identity from Multilingual Social Media Data. *Transactions in GIS* (Under review).

Niu, L.; Xing, X.; Liu, Z.; Yao, E. (2025). A Traffic Assignment Model Balancing User Preference and System Optimum. In preparation.

Zhu, H.; Zhang, W.; Huang, N.; Li, B.; **Niu, L.**; ...; Liu, X. (2024). PlanGPT: Enhancing Urban Planning with Tailored Language Model and Efficient Retrieval. *arXiv* (Preprint). <https://doi.org/10.48550/arXiv.2402.19273>